# SAINet: Scale-Aware Adaptive Interactive Semantic Segmentation Network for Remote Sensing Scenarios

Shuaiqi Liu, *Member, IEEE*, Zhonghao Deng, Jiahui Cao, Bing Li, Yudong Zhang, *Senior Member, IEEE*, and Hui Wang

***Abstract*—In the semantic segmentation of remote sensing images (RSIs), heterogeneous modalities often exhibit distribution discrepancies and noise interference. How to effectively mitigate modality gaps and reduce noise while fully exploiting the complementary information across multimodal data remains a challenging problem. To tackle this issue, a scale-aware adaptive interactive network (SAINet) was proposed for the semantic segmentation of remote sensing scenes. First, SAINet employs an adaptive feature interaction (AFI) module to align features from optical RSIs and digital surface models (DSMs) at both spatial and channel levels, thereby alleviating cross-modal inconsistency and enhancing the complementary utilization of multimodal information. Second, by introducing hierarchical contextual semantic representations, SAINet constructs a multiscale context aggregation (MSCA) module, which strengthens the model's capability to perceive objects at different scales as well as complex land-cover distributions. Furthermore, a symmetric dilated spatial mixer (SDSM) is designed to effectively balance local and global dependency modeling, while maintaining favorable computational efficiency and stability when processing high-resolution inputs. Finally, comprehensive experiments are performed on two widely used segmentation datasets, namely, Vaihingen and Potsdam, and the obtained results indicate that SAINet delivers superior segmentation performance.**

***Index Terms*—Adaptive feature interaction (AFI), cross-modal information complementarity, multiscale representation, remote sensing image (RSI) segmentation, semantic segmentation.**

## I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) is one of the core tasks in the extraction and analysis of remote sensing information. Its goal is to divide the pixels in RSIs into different semantic categories, thereby achieving fine recognition of ground objects [1]. RSIs are typically acquired from satellites, uncrewed aerial vehicles (UAVs), or aerial photography platforms, and are characterized by high spatial resolution and complex backgrounds, which makes semantic segmentation particularly challenging. Semantic segmentation of RSIs plays a transformative role in the characterization of terrestrial surfaces [2], [3], [4], land-use monitoring [5], [6], [7], urban planning [8], [9], [10], ecological and environmental protection [11], disaster assessment [12], and resource management, among others. High-precision RSI segmentation can provide reliable data support for decision-making processes. However, with the rapid advancement of remote sensing technologies, semantic segmentation of RSIs in complex scenarios faces increasingly severe challenges. Specifically, optical RSIs are highly susceptible to cloud occlusion and adverse weather conditions during semantic segmentation [13]. In addition, a single type of remote sensing data source often exhibits inherent limitations in information representation when dealing with diverse ground-object segmentation tasks [14]. When unimodal information fails to meet the requirements of information extraction, multimodal data fusion becomes an effective strategy for improving segmentation accuracy [15]. With RGB and digital surface models (DSMs) serving as representative modalities, the introduction of elevation information in DSM provides effective geometric constraints for precise object boundary extraction [16]. Therefore, by jointly exploiting the spectral–textural information from optical RSIs and the vertical spatial characteristics from DSM data, multimodal fusion strategies can fully leverage the complementary advantages across different data sources, thereby significantly enhancing semantic segmentation performance [17], [18]. The observed accuracy gains can be largely credited to the tight coupling between semantic land-cover entities and altimetric characteristics. By leveraging height-related variables, the model can more accurately distinguish between classes that exhibit similar spectral profiles but reside at different elevations. The geometric structures embedded in elevation data not only compensate for the limitations of spectral information under the "same-spectrum-for-different-objects" phenomenon, but also provide an important spatial discriminative criterion for decoupling semantic attributes [19]. Consequently, in this

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article, we construct a multimodal segmentation model that takes RGB and DSM as inputs, extracts features from both modalities, and enables their interaction to obtain more discriminative semantic representations.

Although multimodal fusion strategies can improve segmentation accuracy, efficiently integrating heterogeneous modal features remains a significant challenge. First, optical imagery and elevation data differ fundamentally in their imaging mechanisms, and the absence of proper feature alignment often introduces severe semantic noise. Second, ground objects in RSIs show significant scale differences, spanning from small vehicles to large buildings, which demands the model to possess strong multiscale contextual representation capabilities. Early image segmentation approaches typically adopted pixel-wise classification strategies, where feature vectors of individual pixels were directly extracted and channeled into a multilayer perceptron (MLP) to undergo nonlinear feature mapping and classification, after which the final prediction was obtained by simply aggregating pixel-level results [20]. Although these approaches are lightweight and computationally efficient, they neglect the contextual information in spatial neighborhoods as well as the structural correlations across different modalities. To overcome these limitations, convolutional neural networks (CNNs) were introduced into image segmentation tasks. CNNs are capable of effectively modeling local spatial structures and have demonstrated superior performance in semantic segmentation [21], [22]. However, CNNs are limited by a fixed receptive field and have obvious shortcomings in long-distance dependency modeling and global context relationship capture. To tackle these problems, Transformers have been gradually introduced into semantic segmentation tasks [23]. Benefiting from their ability to establish feature dependencies over a global scope, Transformers significantly enhance long-range context modeling and improve semantic consistency in large-scale scenes. Nevertheless, existing studies indicate that Transformers still face challenges in the semantic segmentation of RSIs. In particular, the self-attention mechanism lacks explicit local spatial modeling, which often leads to insufficient structural delineation in boundary regions. This has led to the widespread adoption of hybrid CNN-Transformer architectures as the mainstream solution for semantic segmentation of RSIs. The core concept behind these models is the synergistic fusion of two complementary paradigms: the proficiency of CNNs in capturing intricate local patterns and the strength of Transformers in modeling global contextual information. This integration helps to understand complex remote sensing scenes more comprehensively, thereby driving significant improvements in segmentation performance. Following this design philosophy, in this article, we also adopt a hybrid architecture and construct a symmetric dilated spatial mixer (SDSM) to effectively balance local and global dependency modeling.

In response to the rapid evolution of deep learning and multimodal remote sensing segmentation techniques, a series of feature fusion strategies has been developed. These are generally distinguished according to the specific architectural phase where data integration is executed, resulting in a classification of initial-level, mid-level, and decision-level fusion strategies. However, the initial-level strategy, which integrates data at the input level, presents a core challenge: it demands exceptionally precise alignment of different data. Failure to achieve this makes the model particularly susceptible to introducing noise, thereby compromising subsequent feature extraction [24]. In contrast, the decision-level strategy integrates different modalities only at the high-level decision stage, which often makes it difficult to fully exploit the deep correlations among heterogeneous modal features [25]. Compared with these approaches, the mid-level strategy performs multistage cross-modal interactions, enabling the modeling of complementary relationships between different modalities across multiple semantic levels. Moreover, by incorporating multiscale features, mid-level fusion progressively enhances the representation capability for complex scenes, thereby achieving better semantic consistency [26]. However, existing mid-level fusion approaches are inherently limited by their reliance on simplistic strategies, such as feature summation or gating mechanisms, which are insufficient for capturing long-range cross-modal dependencies across different stages. As a result, their performance remains limited in complex remote sensing scenarios [27], [28]. To address this issue and facilitate more effective feature interaction, in this article, we propose an adaptive feature interaction (AFI) module and a multiscale context aggregation (MSCA) module.

In response to the above challenges, we propose a scale-aware adaptive interactive network (SAINet) for multimodal semantic segmentation of RSIs. SAINet is a multimodal segmentation framework that takes RGB images and DSM data as two input modalities. The architecture of SAINet is designed to progressively refine feature representations. Both branches first leverage a ResNet50 backbone to extract rich local features. To enable effective cross-modal fusion and interaction, these features are subsequently processed by an AFI module. Then, to capture broader contextual relationships, an MSCA module is constructed, which strengthens the network's global semantic understanding. Finally, we introduce an SDSM module to preserve local spatial structures within the Transformer framework while capturing cross-scale information, ultimately improving the segmentation quality of RSIs. Specifically, the main contributions of this article are summarized as follows.

1) To tackle the contradiction between the retention of intricate spatial textures and the capture of extensive semantic dependencies in RSIs, we propose a novel semantic segmentation framework for RSIs—SAINet. It combines the local representation capability of CNN with the global modeling capability of Transformer, effectively balancing fine-grained boundary characterization and large-scale contextual modeling.
2) To overcome the limitations in cross-modal feature interaction for multimodal remote sensing data, we design an AFI module. By introducing dynamic modulation mechanisms at both spatial and channel levels, the AFI module adaptively calibrates and aligns RGB and DSM features, enabling effective suppression of modal noise while facilitating efficient interaction and deep fusion of complementary information during multiscale feature propagation.

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3) To handle the significant scale variations of ground objects in RSIs, we construct the MSCA module. This module models contextual information under different receptive fields through parallel convolutional branches with multiple dilation rates, thereby enhancing multiscale feature representation and effectively improving the discrimination of small objects and large-scale structures in complex backgrounds.
4) To alleviate the insufficient modeling of local spatial structures, we build an SDSM module to strengthen both global and local spatial modeling capabilities. By incorporating local convolutions within the Transformer framework, SDSM enables structured feature mixing, which compensates for the limited spatial modeling ability of global attention mechanisms while reinforcing the deep aggregation of local spatial relationships and multiscale semantic information.

The rest of this article is structured as follows. Section II reviews related work on unimodal and multimodal semantic segmentation methods. Section III introduces the architecture of the proposed SAINet. Section IV presents experimental evaluations and performance analysis. Finally, Section V concludes this article.

## II. Related Work

### A. Unimodal Segmentation Methodologies

Compared with natural images, RSIs typically exhibit more complex spatial structures, larger scene scales, and more pronounced variations in object scales [29]. Early segmentation methods mainly relied on handcrafted features combined with traditional classifiers for pixel-wise prediction. Due to the limited representational capacity of such features, these approaches generally showed weak generalization performance in complex scenes [30]. Driven by deep learning, the field of unimodal remote sensing semantic segmentation has been successively shaped by CNNs and, more recently, by Transformers, marking two different technological directions. Representative CNN-based unimodal segmentation models include the U-Net [31], SegNet [32], and the DeepLab series [33], [34], [35]. Typical Transformer-based unimodal segmentation models include Swin-Unet [36], SegFormer [37], LVT [38], and OneFormer [39]. Although the above-mentioned deep learning-based image segmentation models have achieved promising segmentation results, several challenges still remain.

To tackle the prevalent issues of scale variation and complex backgrounds in RSIs, MSCA mechanisms have been extensively introduced to enlarge the receptive field and enhance feature representation capability. For example, remote sensing segmentation models based on PSPNet [40] have demonstrated effectiveness in capturing multiscale contextual information. Building upon these designs, models such as ResUNet-a [41] and TreeUNet [42] further integrate residual connections, parallel dilated convolutions, and pyramid structures within the U-Net framework to improve multiscale feature fusion. However, existing multiscale modules often obtain large receptive fields at the cost of local spatial precision and exhibit insufficient discriminability for ground objects under complex backgrounds. To mitigate these limitations, we construct an MSCA module. By employing parallel convolutions with varied dilation rates, it captures contextual information across diverse receptive fields, which significantly strengthens the model's multiscale feature representation.

In recent years, the segmentation models based on the hybrid architecture of CNN and Transformer have become mainstream. For example, Li et al. [43] designed a bilateral attention framework featuring spatial and context paths, achieving impressive results; Wang et al. [44] explored a hybrid architecture, pairing a convolutional encoding backbone with a Transformer-driven decoding module to achieve balanced local–global modeling. This approach achieves collaborative modeling of local and global information with high computational efficiency, leading to superior boundary segmentation in RSIs. Yao et al. [45] injected global information from Transformers into CNNs to obtain a broader global view, thereby enhancing the modeling capability for complex scenes. Although these methods integrate local and global features to some extent, existing hybrid architectures often adopt simplistic fusion strategies during feature interaction, making it difficult to fully preserve structured spatial relationships. This limitation is particularly evident in boundary-sensitive regions of RSIs. Therefore, we build the SDSM module upon a hybrid architecture to compensate for the insufficient spatial modeling ability of global attention mechanisms while strengthening the deep aggregation of local spatial relationships and multiscale semantic information.

### B. Multimodal Segmentation Methodologies

With the proliferation of multimodal data acquisition systems, remote sensing semantic segmentation has rapidly evolved. While unimodal models focus on one source, multimodal strategies aim to jointly utilize complementary information from different data sources for image semantic segmentation. By integrating the spectral richness of optical imagery with the height information from DSM, these methods aim to achieve a more comprehensive and accurate interpretation of ground objects, thereby improving segmentation robustness in complex scenes. It should be noted that different modality combinations usually involve different fusion challenges. For RGB–DSM or RGB–depth data, the auxiliary modality mainly provides elevation or geometric cues, which can help distinguish objects with similar spectral appearances but different height distributions. In contrast, RGB–SAR fusion faces larger discrepancies in imaging mechanisms and feature distributions, since SAR data reflects microwave backscattering characteristics rather than optical spectral information. Therefore, effective multimodal segmentation requires modality-aware feature alignment and noise suppression. For example, FTransDeepLab [46] introduced multimodal feature calibration and cross-attention-based feature fusion mechanisms within the DeepLabv3 + framework, enabling collaborative optimization of global and local information. For RGB–depth-based remote sensing semantic segmentation tasks, DFormerv2 [47] incorporates depth information as a geometric prior into the self-attention mechanism, achieving effective data fusion with relatively low computational cost.

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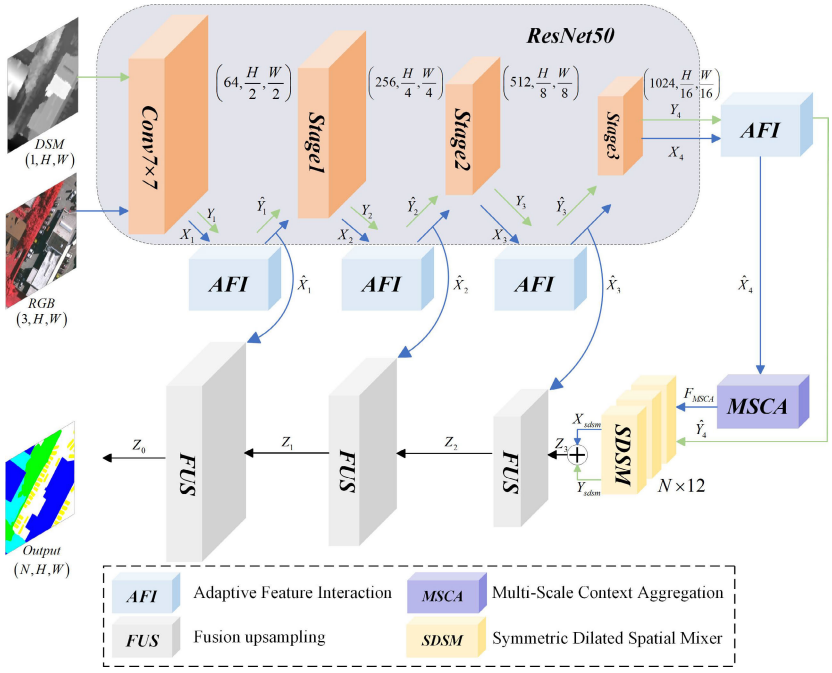



Fig. 1. Overall architecture of SAINet. The inputs of the model are optical (RGB) images and DSM images. The backbone is ResNet50. Subsequently, the AFI module performs cross-modal feature interaction, and the MSCA module further achieves MSCA. Finally, the SDSM module enhances deep aggregation of local spatial relationships and multiscale semantics by introducing local convolution operations into the Transformer framework.

Ni et al. [48] proposed a multimodal segmentation network that dynamically adjusts class-wise interaction weights between modalities to maximize their complementary benefits. Chen et al. [49] designed a dual-branch network based on RGB and SAR data, which improves semantic segmentation by jointly modeling information in both spatial and frequency domains. To simultaneously model local details and global semantic information, Ma et al. [50] proposed FTransUNet, which integrates convolution- and Transformer-based fusion strategies at different network hierarchies. This approach fuses shallow and deep features in a multilevel manner, effectively leveraging CNNs for accurate local detail representation and Transformers for global semantic modeling. Recent studies also find that multistage reconstruction can improve the interpretability and effectiveness of multimodal fusion networks [51], [52].

Although the aforementioned methods improve remote sensing semantic segmentation performance to some extent through multiscale modeling and cross-modal interaction, existing approaches often lack adaptive regulation capabilities across different feature hierarchies. Moreover, attention-based fusion methods frequently introduce feature redundancy, leading to increased computational complexity. Therefore, while fully exploiting multimodal complementary information, how to effectively preserve fine-grained boundaries and local structural features remains a critical challenge in multimodal remote sensing semantic segmentation. To tackle these issues, we propose SAINet, which not only integrates CNNs and Transformers to effectively balance fine-grained boundary delineation and global context modeling, but also constructs an AFI module by introducing dynamic modulation mechanisms at both spatial and channel levels. This design enables adaptive calibration and alignment of RGB and DSM features, thereby facilitating efficient interaction and deep fusion of complementary information during multiscale feature propagation.

## III. Method

In this article, we propose an efficient semantic segmentation framework named SAINet, tailored for multimodal RSIs. As illustrated in Fig. 1, its overall architecture integrates several key components: a backbone network for feature extraction, an AFI module for cross-modal feature fusion and interaction, a MSCA module for capturing multiscale contexts, a SDSM module for reinforcing the deep aggregation of local spatial relationships and multiscale semantic information, and multiple fusion upsampling (FUS) modules for progressive resolution recovery.

In the encoding stage, SAINet takes a three-channel RGB optical image $X \in \mathbb{R}^{3\times H\times W}$ and a single-channel DSM image $Y \in \mathbb{R}^{1\times H\times W}$ as inputs, respectively. A pair of ResNet50 networks with identical architectures is adopted as the backbone networks to extract multiscale information from these diverse inputs. Here, $C$, $H$, and $W$ denote the number of channels, height, and width of the input image, respectively. The four stages of ResNet50 produce feature maps at four different resolutions, denoted $X_1, Y_1 \in \mathbb{R}^{64\times H/2\times W/2}$, $X_2, Y_2 \in \mathbb{R}^{256\times H/4\times W/4}$, $X_3, Y_3 \in \mathbb{R}^{512\times H/8\times W/8}$, and $X_4, Y_4 \in \mathbb{R}^{1024\times H/16\times W/16}$. Subsequently, SAINet feeds the features from the two modalities at each corresponding stage into the AFI module to perform feature interaction, producing aligned representations $\hat{X}_i, \hat{Y}_i \in \mathbb{R}^{C_i\times H_i\times W_i}$. At the deepest semantic level $X_4 \in \mathbb{R}^{1024\times H/16\times W/16}$, to effectively capture regional structures on a cross-scale, SAINet introduces the MSCA module. This

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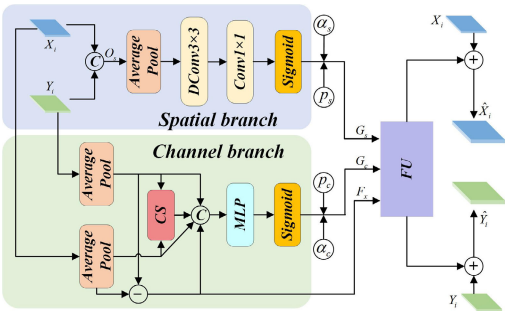



Fig. 2. Architecture of the AFI module, where RGB features and DSM features are taken as inputs to enable cross-modal interaction.

module leverages parallel dilated convolutions with varying dilation factors to model contextual relationships across different receptive fields, thereby effectively capturing cross-scale regional structures. It is worth noting that experimental results demonstrate that applying the MSCA module to DSM features for cross-scale modeling leads to a degradation in network performance. This is because DSM data primarily encode geometric elevation information, for which the most critical features are local height differences and object boundaries. Introducing the MSCA module may introduce irrelevant long-range elevation interference, which in turn blurs originally clear physical boundaries.

### A. AFI Module

To enhance the interaction between the two modal features, we design an AFI module, which consists of average pooling, convolution operations, activation functions, an MLP, and a feature update (FU) module. Fig. 2 exhibits the detailed framework of the AFI module. By harvesting the inherent synergy between different modalities, the AFI adaptively optimizes the feature responses. This interactive mechanism spans both spatial and channel dimensions to preserve structural integrity.

In the specific implementation, the AFI module performs a channel-wise concatenation of the incoming multimodal data to yield the feature $O_s$, which is formulated as

$$O_s = Concat\left(X_i, Y_i\right) \tag{1}$$

where $X_i$ and $Y_i$ denote the RGB and DSM features at the $i$th stage, respectively, and both have $C$ channels. The operator $Concat(\cdot)$ represents channel-wise concatenation. Consequently, $O_s$ denotes the concatenated feature map with channels.

The AFI module first processes the concatenated feature $O_s$ by applying an average-pooling operation, which is followed by a $3 \times 3$ depthwise convolution on the merged feature $O_s$. Subsequently, a $1 \times 1$ convolution combined with a Sigmoid function is employed to generate a spatial gate low-resolution spatial gate $\hat{G}_s \in \mathbb{R}^{B\times 1\times H/2\times W/2}$. Finally, bilinear interpolation is used to restore the spatial gate to the original spatial size, resulting in $G_s \in \mathbb{R}^{B\times 1\times H\times W}$. This gating mechanism characterizes the pixel-wise influence of features from one modality on those of another modality, thereby highlighting spatial regions with complementary information and being particularly adept at capturing fine edge and boundary details in RSIs. The overall computation can be formulated as

$$\hat{G}_s = \sigma\left(Conv_{1\times 1}\left(DConv_{3\times 3}\left(Avg\left(O_s\right)\right)\right)\right) \tag{2}$$

$$G_s = UP\left(\hat{G}_s\right) \tag{3}$$

where $\sigma(\cdot)$ denotes the Sigmoid function, $Conv_{1\times 1}(\cdot)$ represents a $1 \times 1$ convolution operation, $DConv_{3\times 3}(\cdot)$ denotes a $3 \times 3$ depthwise convolution, $Avg(\cdot)$ denotes average pooling with a downsampling ratio of 2, and $UP(\cdot)$ denotes bilinear interpolation function, which performs a $2\times$ upsampling operation on the feature.

Subsequently, the AFI module applies global average pooling to obtain feature representations $X_i$ and $Y_i$ to obtain channel features $X_c$ and $Y_c$, respectively. In Fig. 2, CS denotes the channel-wise cosine decomposition operation. Specifically, $X_c$ and $Y_c$ are first normalized along the channel dimension, and their element-wise product is computed to obtain the channel-wise correlation vector $F_s$. Meanwhile, the absolute difference between $X_c$ and $Y_c$ is calculated as $F_x$. Finally, these features are concatenated to form the fused channel feature $F_t$, that is,

$$F_x = Abs\left(X_c - Y_c\right) \tag{4}$$

$$F_s = \frac{X_c}{\|X_c\|_2} \odot \frac{Y_c}{\|Y_c\|_2} \tag{5}$$

$$F_t = Concat\left(\hat{X}_c, \hat{Y}_c, F_s, F_x\right) \tag{6}$$

where $Abs(\cdot)$ denotes the absolute value operation, and $\odot$ denotes element-wise multiplication. The channel number of $F_x$ is $C$, while the channel number of $F_t$ is $4C$. Subsequently, the AFI module feeds the concatenated feature $F_t$ into an MLP to generate channel-wise attention weights $G_c \in \mathbb{R}^{B\times C\times 1\times 1}$, which are used to selectively enhance channels with strong semantic consistency while suppressing noisy or redundant features. The overall computation process can be expressed as

$$G_c = \sigma\left(Linear\left(ReLU\left(Linear\left(F_t\right)\right)\right)\right) \tag{7}$$

where $Linear(\cdot)$ denotes a linear projection function, and $ReLU(\cdot)$ represents the ReLU function.

To prevent the AFI module from excessively disturbing the feature distribution learned by the backbone network at the early stage of training, two learnable parameters $[\alpha_s, \alpha_c]$ are introduced, both of which are initialized to 0.1. As training proceeds, the model adaptively increases or decreases these two scaling factors, enabling the interaction strength between multimodal features to be progressively enhanced or weakened. In addition, two global weighting coefficients, $p_s$ and $p_c$, are introduced to balance spatial interaction and channel interaction, respectively. Both coefficients are initialized to 0.5 and satisfy the constraint $p_s + p_c = 1$, where $p_s$ and $p_c$ indicate the relative importance of spatial and channel interactions. The FU module is mainly responsible for feature updating, and its specific computation process is given by

$$\hat{F}_{o1} = \left(p_s \cdot \alpha_s \cdot G_s + p_c \cdot \alpha_c \cdot G_c\right) \odot F_x \tag{8}$$

$$\hat{F}_{o2} = \left(p_s \cdot \alpha_s \cdot (1 - G_s) + p_c \cdot \alpha_c \cdot (1 - G_c)\right) \odot F_x \tag{9}$$

where $\odot$ denotes element-wise multiplication.

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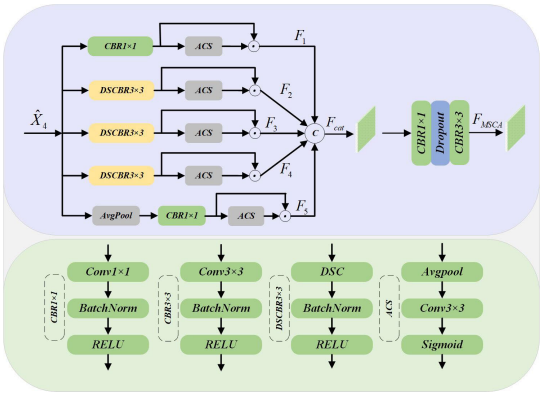



Fig. 3. Architecture of the MSCA module, where parallel multiscale branches and adaptive weighting are used to enhance contextual feature representation.

Finally, the AFI module adds the updated features $\hat{F}_{o1}$ and $\hat{F}_{o2}$ to the corresponding inputs $X_i$ and $Y_i$ of the AFI module, respectively, to obtain the output features $\hat{X}_i$ and $\hat{Y}_i$

$$\hat{X}_i = \hat{F}_{o1} + X_i, \quad \hat{Y}_i = \hat{F}_{o2} + Y_i. \tag{10}$$

The AFI module realizes the bidirectional complementarity and adaptive correction of multimodal features, promoting the gradual alignment of bimodal features into a unified semantic space. It is worth noting that although the AFI module contains multiple learnable weights, it is composed only of a small number of lightweight parameters and simple gating mechanisms, which introduce almost no additional computational overhead. Compared with the relatively low computational cost, the improvements in stability and fusion accuracy brought by these learnable weights are significantly more pronounced.

### B. MSCA Module

To solve the issue that ground objects usually exhibit complex shapes, diverse scales, and significant spatial heterogeneity, we design an MSCA module, which aims to perform multiscale feature enhancement and channel-adaptive calibration before entering deep feature interaction. The MSCA module is designed to enhance deep semantic features by capturing contextual information under different receptive fields through multiscale convolutional structures. By dynamically adjusting feature responses via adaptive weighting mechanisms, MSCA strengthens the representation of objects with diverse scales and improves the discrimination of complex land-cover regions.

Each branch of the MSCA employs a different type of convolution operation to perceive spatial information at varying scales. The detailed network structure of the MSCA module is illustrated in Fig. 3. Specifically, the top branch of the MSCA module utilizes a $1 \times 1$ convolution to perform channel mapping and semantic integration on the input features, thereby obtaining compact global semantic representations. The three intermediate branches employ depth-wise separable convolutions (DSCs) with different dilation rates (5, 9, and 13) to capture multiscale contextual features and enlarge the effective receptive field. The bottom branch extracts global features through average pooling, followed by a $1 \times 1$ convolution, and further incorporates adaptive weights to achieve global semantic compensation.

For a given input feature $\hat{X}_4 \in \mathbb{R}^{B\times C\times H\times W}$, the output of the $i$th branch in the MSCA module can be expressed as

$$F_1 = ACS\left(CBR_{1\times1}\left(\hat{X}_4\right)\right), \quad i = 1 \tag{11}$$

$$F_i = ACS\left(DSCBR_{3\times3}\left(\hat{X}_4\right)\right), \quad i = 2,3,4 \tag{12}$$

$$F_i = ACS\left(CBR_{1\times1}\left(Avg\left(\hat{X}_4\right)\right)\right), \quad i = 5 \tag{13}$$

where $CBR_{1\times1}(\cdot)$ denotes the mapping function of a module composed of a convolution with the kernel size $1 \times 1$, batch normalization, and a ReLU function. $DSCBR_{3\times3}(\cdot)$ represents the mapping function of a module consisting of a DSC operation with the kernel size $3 \times 3$, batch normalization, and a ReLU function. $ACS(\cdot)$ denotes the adaptive weight computation operation, which is composed of global average pooling, a convolution with the kernel size $1\times1$, and a Sigmoid function to generate the adaptive weight map $A_i \in [0,1]$. Subsequently, the MSCA module performs element-wise multiplication between the adaptive weights and the feature $F_i$ to achieve adaptive feature reweighting

$$F_i' = A_i \odot F_i. \tag{14}$$

Finally, the outputs of each branch are concatenated in the channel dimension to obtain $F_{cat}$

$$F_{cat} = Concat\left(F_1', F_2', F_3', F_4', F_5'\right). \tag{15}$$

Then, the MSCA module obtains the final multiscale enhanced feature $F_{MSCA}$ from $F_{cat}$ through convolution and regularization operations. Through the promotion of cross-scale feature synergy and the suppression of overfitting, the MSCA module enhances the generalization capacity of the model when dealing with complex situations.

### C. Symmetric Dilated Spatial Mixer

Multihead self-attention (MHSA) is capable of effectively modeling global dependency relationships; however, it still suffers from certain limitations in dense prediction tasks. On the one hand, attention mechanisms primarily rely on global feature similarity for representation learning, which makes them less sensitive to local spatial structures and thus insufficient for accurately modeling fine-grained boundary and texture details. On the other hand, the global weighting property of MHSA tends to over-smooth features when processing high-resolution features, which is unfavorable for preserving subtle spatial variations. These limitations are particularly pronounced in RSI segmentation tasks, where targets usually exhibit complex local structures and multiscale spatial distributions.

To better handle the above problems, we construct the SDSM module, in which the MHSA in a standard Transformer is replaced with a spatial reshape enhancement (SRE) module. The network architectures of the standard Transformer and the proposed SDSM are illustrated in Fig. 4. In the SDSM module, the SRE module leverages convolutional operations to enable feature interactions in the spatial domain, thereby

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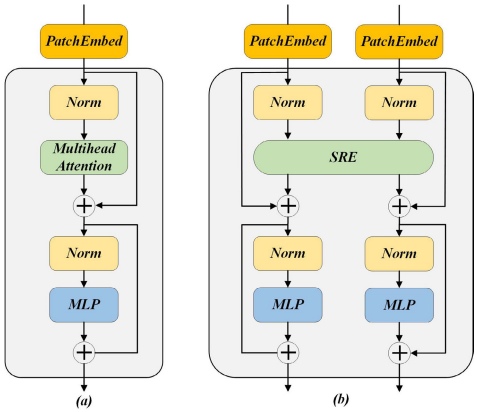



Fig. 4. Network structures of the proposed SDSM and the standard transformer. (a) Standard transformer architecture. (b) Architecture of the SDSM module.

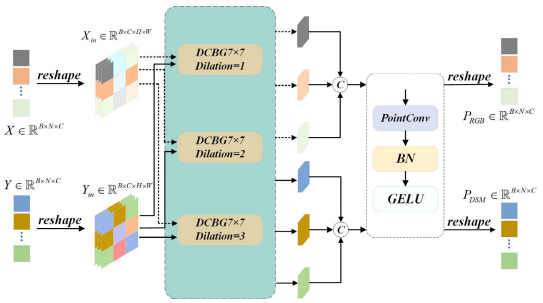



Fig. 5. Architecture of the SRE module. The input features are first reshaped into 2-D feature maps, then processed through parallel DSCs, pointwise convolutions, and activation operations, and reshaped into sequence features to enhance local spatial modeling.

more effectively capturing local spatial patterns and multiscale contextual information.

The SRE module consists of multiple depthwise convolution branches with multiple dilation rates $d_i$ to gather multiscale contextual information. As depicted in the architectural diagram (Fig. 5), the SRE module performs identical operations on the RGB branch and the DSM branch features. Taking the RGB branch as an example, for a given input feature sequence, it is first transformed into a feature $X_{in} \in \mathbb{R}^{B\times C\times H\times W}$. The SRE module employs depthwise convolutions with different $d_i$ to capture multiscale context from $X_i$. The output of the depthwise convolution with $d_i$ can be expressed as

$$P_i = GELU\left(BN\left(DConv_{k,d}\left(X_{in}\right)\right)\right) \tag{16}$$

where $k$ denotes the convolution kernel size, $d_i \in \{1,2,3\}$ represents the dilation rate, and $GELU(\cdot)$ denotes the GELU function. Subsequently, the SRE module concatenates the outputs of the three branches along the channel dimension to obtain the feature $P'$, which is then processed by a pointwise convolution, batch normalization, and a GELU function to generate the feature $P_{RGB}$

$$P_{RGB} = GELU\left(BN\left(\mathrm{Conv}_{1\times1}\left(P'\right)\right)\right). \tag{17}$$

Similarly, the feature obtained from the DSM branch through the SRE module is denoted as $P_{DSM}$. Finally, the SRE module reshapes $P_{RGB}$ and $P_{DSM}$ into sequence form and feeds them into the subsequent SDSM module for further processing.

By embedding a multibranch depthwise convolution structure, the SRE module is capable of modeling rich contextual relationships within local regions while simultaneously preserving fine-grained details and regional semantic features. Compared with MHSA, the feature interaction mechanism of the SRE module is better for RSI segmentation tasks and maintains more stable spatial feature representations in complex scenarios. Therefore, the SDSM can effectively enhance the model's spatial structural modeling capability. Finally, the FUS module performs multimodal feature fusion on the features obtained after passing through 12 stacked SDSM modules.

### D. FUS Module and Loss Function

Let the outputs of the last SDSM module be $X_{sdsm}$ and $Y_{sdsm}$. Before performing multimodal fusion, $X_{sdsm}$ and $Y_{sdsm}$ are first summed element-wise to obtain the feature $Z_i$, which contains low-resolution information from both RGB and DSM modalities. To obtain high-resolution multimodal fusion information, a FUS module is constructed. The inputs of the FUS module include the feature $Z_i$ and the output $\hat{X}_i$ from the corresponding AFI module at the same level. In the FUS module, $Z_i$ is first upsampled to the same spatial resolution as $\hat{X}_i$. Following this, the pair of features is concatenated along the channel dimension $\hat{X}_i$, and a convolution with kernel size $3 \times 3$ is subsequently applied to compress the channel depth, yielding the output feature $Z_{i-1}$:

$$Z_{t-1} = Conv_{3\times3}\left(\mathrm{Concat}\left(UP\left(Z_t\right), \hat{X}_{t1}\right)\right) \tag{18}$$

where $UP(\cdot)$ denotes the bilinear interpolation function, which performs a $2\times$ upsampling operation on the feature.

To guide parameter optimization and updating during training, a composite loss function $\mathcal{L}_{total}$ is adopted, which is the weighting of the four-stage loss functions $\mathcal{L}_t$, that is,

$$\mathcal{L}_{total} = \mathcal{L}_4 + \alpha_3 * \mathcal{L}_3 + \alpha_2 * \mathcal{L}_2 + \alpha_1 * \mathcal{L}_1 \tag{19}$$

where $\mathcal{L}_1$ denotes the loss at the input stage of the first FUS module, and $\mathcal{L}_t$ $(t = 2,3,4)$ represents the loss at the output stage of the corresponding FUS module. To ensure the accuracy of pixel prediction at the bottom layer and the fitting ability to the target shape and boundary, each stage loss $\mathcal{L}_t$ consists of the cross-entropy loss $\mathcal{L}_{ce}$ and dice loss $\mathcal{L}_{dice}$, that is,

$$\mathcal{L}_t = \mathcal{L}_{ce}^t + \mathcal{L}_{dice}^t. \tag{20}$$

The cross-entropy loss and dice loss are, respectively, defined as

$$\mathcal{L}_{ce}^t = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{t(n)} \tag{21}$$

$$\mathcal{L}_{dice}^t = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{y_k^{(n)}\hat{y}_k^{t(n)}}{y_k^{(n)} + \hat{y}_k^{t(n)}} \tag{22}$$

where $\mathcal{L}_{ce}^t$ is adopted to evaluate the difference between the predicted class probabilities $\hat{y}_k^{t(n)}$ of $N$ sample over $K$ categories (which are obtained by upsampling $Z_i$ to the same size as the ground truth (GT) and then computing the probability statistics) and the GT labels $y_k^{(n)}$. The $\mathcal{L}_{dice}^t$ focuses on high-confidence predictions and helps enhance model performance in scenarios with class imbalance.

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE VAIHINGEN DATASET

| Type | Method | Year | IoU(%) | | | | | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|---|---|---|---|
| | | | Imp. | Building | Low. | Trees | Car | | |
| Unimodal | TransUnet [53] | 2021 | 83.10 | 89.25 | 65.32 | 82.70 | 70.45 | 87.46 | 78.16 |
| | ABCNet [43] | 2021 | 82.45 | 90.13 | 64.59 | 81.95 | 61.83 | 85.65 | 76.19 |
| | UnetFormer [44] | 2022 | 85.76 | 92.78 | 67.33 | 83.22 | 80.75 | 89.85 | 81.97 |
| | CMTFNet [54] | 2023 | 86.09 | 94.09 | 66.64 | 82.19 | 81.60 | 89.91 | 82.12 |
| | SSNet [45] | 2024 | 86.08 | 93.79 | 67.40 | 83.60 | 78.73 | 89.87 | 81.92 |
| | WCMamba [55] | 2025 | 86.63 | 92.01 | **73.20** | 81.63 | 80.56 | 90.45 | 82.81 |
| Multimodal | CMFNet [56] | 2022 | 84.63 | 91.99 | 65.78 | 82.34 | 75.98 | 88.71 | 80.15 |
| | ASMFNet [57] | 2024 | 85.69 | 88.30 | 66.63 | 80.01 | 81.86 | 88.99 | 80.50 |
| | FTransUnet [50] | 2024 | 87.69 | **95.28** | 69.02 | 84.89 | 81.69 | 90.89 | 83.71 |
| | AFMamba [58] | 2025 | - | - | - | - | - | 90.56 | 83.12 |
| | FDMF-Net [49] | 2025 | 85.92 | 92.99 | 68.50 | 84.16 | 84.66 | 90.64 | 83.25 |
| | Ours | - | **87.76** | 94.51 | 70.14 | **84.96** | **86.66** | **91.57** | **84.81** |

## IV. EXPERIMENTS AND ANALYSIS

### A. Experimental Settings

To rigorously assess the segmentation performance of the proposed SAINet, all experiments were conducted on a dedicated workstation equipped with the Windows 10 operating system. The hardware backbone of this study was an NVIDIA GeForce RTX 3090 GPU featuring 24 GB of RAM, providing sufficient computational capacity for high-resolution processing. On the software side, the entire architecture was implemented using the PyTorch 2.0.0 deep learning framework. In our implementation, the backbone-related parameters were initialized using publicly available pretrained weights. For model optimization, in this article, we employed the AdamW optimizer, initializing the training process with a base learning rate and a weight decay coefficient of 0.01 to facilitate stable convergence and mitigate potential overfitting. To balance training efficiency with memory constraints, the batch size was set to 20. The training procedure spanned 40 complete epochs, after which the model weights were frozen for subsequent performance evaluation. The effectiveness of SAINet was comprehensively validated on two widely recognized benchmarks: the ISPRS Vaihingen and Potsdam datasets. Given the substantial dimensions of the original aerial imagery, all images were partitioned into smaller patches of $256 \times 256$ pixels for both the training and test phases. To further enhance the robustness and generalization capabilities of the model, several stochastic data augmentation strategies were integrated into the training pipeline. Specifically, input patches underwent random horizontal flipping and vertical flipping.

The Vaihingen dataset comprises 33 images of varying sizes, with an average spatial resolution of approximately $2496 \times 2064$ pixels. Each image was meticulously extracted from a larger orthophoto mosaic, excluding regions with invalid data. Among these, 16 images are provided with pixel-level GT annotations covering six semantic classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. In our experimental setup, we utilized 12 images for model training and reserved the remaining 4 for testing. Similarly, the Potsdam dataset offers a larger scale representation of urban scenes, containing 24 images with a fixed dimension of $6000 \times 6000$ pixels. Sharing the same semantic class labels as Vaihingen, the Potsdam dataset was partitioned into 18 tiles for training and 6 tiles for independent performance evaluation.

To thoroughly validate the superiority of SAINet on both datasets, this article compared it with eleven advanced methods, covering both unimodal and multimodal segmentation methods. The unimodal models include ABCNet [43], TransUNet [53], UNetFormer [44], CMTFNet [54], SSNet [45], and WCMamba [55]. The multimodal models include CMFNet [56], ASMFNet [57], FTransUNet [50], AFMamba [58], and FDMF-Net [49].

### B. Evaluation Metrics

To quantitatively verify the effectiveness of the model in the segmentation task, we adopt two objective metrics, namely, the mean $F1$-score (mF1) and the mean intersection over union (mIoU). The $F1$-score and IoU are computed as follows:

$$F_1 = 2 \times \frac{Q_c \times R_c}{Q_c + R_c} \tag{23}$$

$$IoU = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{24}$$

where $c$ denotes a specific category, and $TP_c$, $FP_c$, and $FN_c$ represent the true positives, false positives, and false negatives of class $c$, respectively. $Q_c$ and $R_c$ are defined as

$$Q_c = \frac{TP_c}{TP_c + FP_c} \tag{25}$$

$$R_c = \frac{TP_c}{TP_c + FN_c}. \tag{26}$$

According to the above definitions, after obtaining the $F_1$ and $IoU$ values for each foreground class, the average values over all foreground classes are calculated to obtain the model's mF1 and mIoU.

### C. Experimental Results on the Vaihingen Dataset

The quantitative comparison results of different image segmentation methods on the Vaihingen dataset are reported in Table I, where the segmentation evaluation metrics for each individual class are $IoU$. In the table, bold values represent

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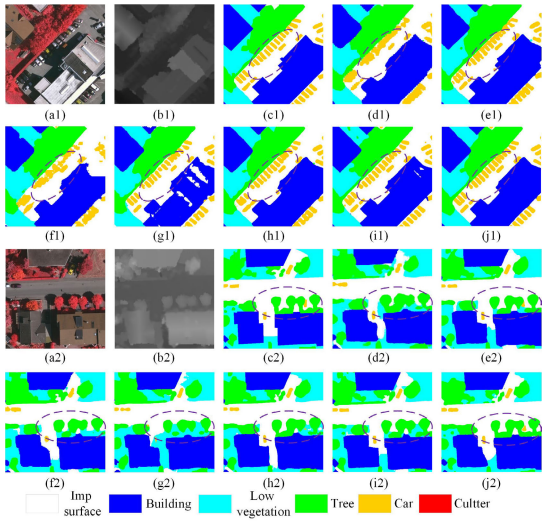



Fig. 6. Qualitative comparison of segmentation results on the Vaihingen dataset. (a1) and (a2) RGB images, (b1) and (b2) DSM images, (c1) and (c2) GT, (d1) and (d2) ABCNet, (e1) and (e2) UNetFormer, (f1) and (f2) ASMFNet, (g1) and (g2) CMFNet, (h1) and (h2) FTransUNet, (i1) and (i2) FDMF-Net, and (j1) and (j2) SAINet.

the best performance, while underlined values correspond to the second-best results. As shown in Table I, among the unimodal methods, WCMamba achieves the highest overall performance, particularly excelling in the low vegetation category, where it attains an IoU of 73.20%. Among the multimodality methods, CMFNet employs a cross-modal multiscale Transformer-based fusion strategy and achieves an mIoU of 80.15%. ASMFNet further enhances cross-modal feature representation through adjacent-scale attention and modality-adaptive fusion, resulting in an overall mIoU of 80.50%. FTransUNet integrates CNNs with an improved Transformer-based fusion module and demonstrates strong performance in the building category, achieving an overall mIoU of 83.71%. In contrast, the proposed method exhibits more balanced performance across all categories, with significant improvements in the tree and car classes. It achieves the highest overall performance among all compared methods, with an mIoU of 84.81% and an mF1 score of 91.57%. These results demonstrate that, by incorporating the AFI, MSCA, and SDSM modules, the proposed model effectively enhances its capability to perceive multiscale targets and complex land-cover distributions, enables sufficient interaction between multimodal information, and achieves a better balance between local and global dependency modeling, thereby significantly improving the robustness of the segmentation performance.

To provide a more intuitive comparison of the segmentation performance, two images were randomly selected from the Vaihingen dataset. The corresponding segmentation results produced by ABCNet, UNetFormer, ASMFNet, CMFNet, FTransUNet, FDMF-Net, and the proposed SAINet are presented in Fig. 6. In the visualization, the category labeled as clutter is regarded as background or a low-density class. RSIs usually have higher resolution than natural images and can present detailed information about ground objects. However, the high resolution also introduces abundant fine details and noise, especially in complex terrains or multiobject scenes, making accurate target segmentation more challenging. To clearly highlight the differences among various methods, regions with significant discrepancies in segmentation results are marked with purple dashed ellipses. As shown in the first row of Fig. 6, buildings often exhibit complex shapes, varying heights, and overlapping features. Their surfaces may resemble roads, making it difficult to accurately delineate building boundaries and increasing the likelihood of misclassification as impervious surfaces. The car in the image is small and difficult to accurately mark, as illustrated in the third row of Fig. 6. It is evident that the proposed SAINet can better address these challenges. As observed in Fig. 6, SAINet effectively models fine-grained targets at both the pixel level and the category level, achieving superior performance across objects of different scales. In particular, for small objects such as cars, SAINet is able to produce clear and precise object boundaries. The strong adaptability of SAINet across diverse scenes can be attributed to the MSCA and AFI modules. Specifically, the MSCA module enhances category representation through effective aggregation of multiscale contextual information, while the AFI module introduces dynamic modulation mechanisms at both spatial and channel levels to align RGB and DSM features, enabling efficient interaction and deep fusion of complementary information during multiscale feature propagation.

### D. Experimental Results on the Potsdam Dataset

To validate the effectiveness of the proposed SAINet on the Potsdam dataset, we further compared it with the aforementioned 11 advanced methods. The quantitative results are summarized in Table II, where the evaluation metric for each individual class is *IoU*. In the table, bold values represent the best performance, while underlined values correspond to the second-best results. As indicated in Table II, among the unimodal methods, WCMamba achieves the highest overall performance, particularly excelling in the low vegetation and tree classes, with IoU values of 77.64% and 79.88%, respectively. Among the multimodal methods, CMFNet employs a cross-modal multiscale Transformer fusion strategy and attains the second-best performance in the building category, with an IoU of 95.67%. FTransUNet integrates CNN and an improved Transformer fusion module, achieving the second-best IoU of 88.39% in the impervious surface category. FDMF-Net introduces a frequency-domain decoupling network and demonstrates strong performance on impervious surfaces, reaching an overall mIoU of 86.63%. In comparison, the proposed method achieves balanced and superior performance across all categories. Notably, significant improvements are observed in the building, low vegetation, tree, and car categories. The proposed SAINet attains an mIoU of 87.67% and an mF1 score of 93.28%, delivering the best performance.

To offer a more intuitive visualization of the segmentation performance, two samples were randomly chosen from the Potsdam dataset. The corresponding segmentation results produced by ABCNet, UNetFormer, ASMFNet, CMFNet, FTransUNet, FDMF-Net, and the proposed SAINet are illustrated in Fig. 7. As depicted in Fig. 7, buildings in this

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE POTSDAM DATASET

| Type | Method | Year | IoU (%) | | | | | mF1 (%) | mIoU (%) |
|---|---|---|---|---|---|---|---|---|---|
| | | | Imp. | Building | Low. | Trees | Car | | |
| Unimodal | TransUnet [53] | 2021 | 85.28 | 92.22 | 73.99 | 74.33 | 90.71 | 90.69 | 83.51 |
| | ABCNet [43] | 2021 | 81.90 | 89.62 | 69.31 | 66.12 | 85.70 | 87.67 | 78.51 |
| | UnetFormer [44] | 2022 | 87.42 | 91.80 | 75.54 | 77.67 | 90.95 | 91.55 | 84.67 |
| | CMTFNet [54] | 2023 | 85.97 | 93.82 | 75.90 | 76.91 | 92.51 | 91.73 | 85.02 |
| | SSNet [45] | 2024 | 85.16 | 93.38 | 74.40 | 75.33 | 91.91 | 91.12 | 84.04 |
| | WCMamba [55] | 2025 | 87.90 | 94.45 | 77.64 | 79.88 | 92.82 | 92.64 | 86.54 |
| Multimodal | CMFNet [56] | 2022 | 86.59 | 95.67 | 75.16 | 76.37 | 91.98 | 92.10 | 85.15 |
| | ASMFNet [57] | 2024 | 85.60 | 93.56 | 75.64 | 75.73 | 91.02 | 91.24 | 84.31 |
| | FTransUnet [50] | 2024 | 88.39 | 95.24 | 77.41 | 77.59 | 92.07 | 92.45 | 86.14 |
| | AFMamba [58] | 2025 | – | – | – | – | – | 92.23 | 86.14 |
| | FDMF-Net [49] | 2025 | **88.58** | 95.17 | 77.38 | 77.86 | 94.18 | 92.65 | 86.63 |
| | Ours | – | 88.05 | **96.25** | **79.75** | **79.93** | **94.37** | **93.28** | **87.67** |

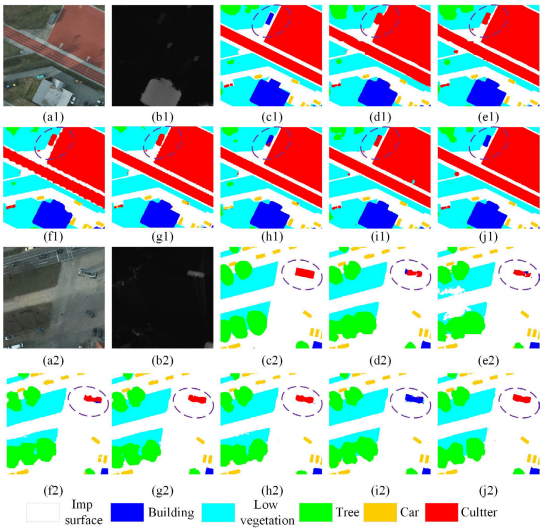



Fig. 7. Qualitative comparison of segmentation results on the Potsdam dataset. (a1) and (a2) RGB images, (b1) and (b2) DSM images, (c1) and (c2) GT, (d1) and (d2) ABCNet, (e1) and (e2) UNetFormer, (f1) and (f2) ASMFNet, (g1) and (g2) CMFNet, (h1) and (h2) FTransUNet, (i1) and (i2) FDMF-Net, and (j1) and (j2) SAINet.

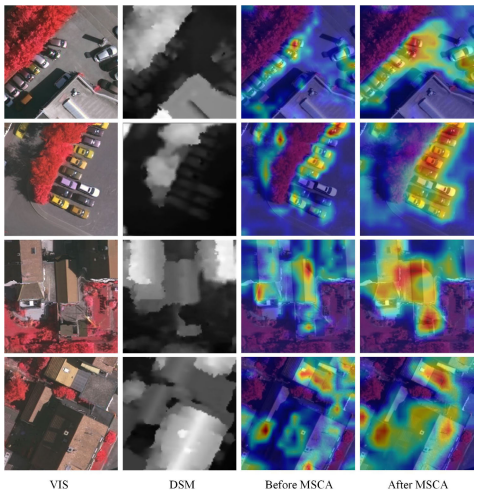



Fig. 8. Visualization results of heatmaps before and after the MSCA module.

dataset exhibit complex shapes, varying heights, and surface appearances similar to clutter, which increases the likelihood of misclassification, as shown in the first and third rows of Fig. 7. To clearly highlight the differences among the compared methods, regions with significant segmentation discrepancies are marked using purple dashed ellipses. As can be seen from Fig. 7, SAINet achieves a notable improvement in distinguishing between clutter and building categories. Specifically, in the circled regions of Fig. 7(a1)–(j1), ABCNet, UNetFormer, ASMFNet, CMFNet, and FDMF-Net misclassify buildings as clutter. Although FTransUNet correctly segments most building regions, some portions are still incorrectly labeled as clutter. In contrast, SAINet successfully achieves accurate semantic segmentation of the circled building regions. This indicates that the proposed MSCA module effectively improves the model's discriminative ability for both small objects and large-scale structures under complex background conditions. Similarly, in the circled regions of Fig. 7(a2)–(j2), ABCNet, UNetFormer, and ASMFNet incorrectly classify clutter as buildings, while FDMF-Net misclassifies most clutter regions as buildings. Although CMFNet, FTransUNet, and SAINet generate correct semantic predictions in these regions, the segmentation produced by SAINet is visually closest to the GT. These results consistently demonstrate the robustness of SAINet, confirming its superior performance in handling segmentation tasks within complex and heterogeneous scenes.

### E. Ablation Study

Ablation experiments were conducted on the Vaihingen dataset to evaluate the effectiveness of each component of the proposed model. The experimental results are shown in Table III. The ablation configurations constructed in this study are as follows: Case 1 as baseline, includes only the backbone network; Case 2 adds the AFI module to baseline; Case 3 adds the MSCA module to baseline; Case 4 adds the SDSM module to baseline; Case 5 adds both the AFI and MSCA modules to baseline; Case 6 adds both the MSCA and SDSM modules to baseline; Case 7 adds both the AFI and SDSM modules to baseline; and Case 8 adds the AFI, MSCA, and SDSM modules to baseline, which is the proposed SAINet model.

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TABLE III
ABLATION STUDY OF AFI, MSCA, AND SDSM ON THE VAIHINGEN DATASET

| Method | Module | | | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|
| | AFI | MSCA | SDSM | | |
| Case1 | | | | 89.93 | 82.10 |
| Case2 | ✓ | | | 91.02 | 84.05 |
| Case3 | | ✓ | | 91.01 | 83.66 |
| Case4 | | | ✓ | 91.26 | 84.32 |
| Case5 | ✓ | ✓ | | 91.25 | 84.22 |
| Case6 | | ✓ | ✓ | 90.53 | 83.30 |
| Case7 | ✓ | | ✓ | 91.29 | 84.43 |
| Case8 | ✓ | ✓ | ✓ | **91.57** | **84.81** |

As shown in Table III, Case 1 yields the worst performance, with an mIoU of 82.10%. Case 2 improves the mIoU to 84.05% by introducing the AFI module, indicating that cross-modal interaction enables complementary information exchange between the two modalities and enhances segmentation performance. Case 3 achieves an mIoU of 83.66% after incorporating the MSCA module, illustrating the benefit of multiscale contextual enhancement for remote sensing semantic segmentation. Case 4 improves the mIoU to 84.32% by introducing the SDSM module, validating the performance gains brought by local feature mixing. Case 5 combines Case 2 and Case 3, further improving segmentation performance to an mIoU of 84.22%, indicating that supplementing multiscale information under modality alignment can enhance semantic segmentation performance. Case 6 combines the added modules in Case 3 and Case 4, but its performance (mIoU of 83.30%) is lower than that of Case 3 and Case 4, suggesting that without the AFI module for cross-modal alignment, the combination of MSCA and SDSM may introduce feature redundancy and over-smoothing, partially offsetting the performance gains. Case 7 combines the added modules in Case 2 and Case 4 and achieves a slight improvement, with an mIoU of 84.43%. This indicates that the cross-modal feature interaction and the local spatial feature mixture have a certain degree of complementarity. AFI module helps to enhance the information fusion between modalities, while SDSM can further strengthen the expression of local details, thereby achieving a more stable performance improvement. Case 8 combines all the modules in Cases 2–4, i.e., the proposed SAINet, and achieves the best mIoU and mF1 values. This is only possible when all modules are jointly incorporated, allowing the model to effectively exploit cross-modal complementarity, extract salient multiscale object features, and strengthen the deep aggregation of local spatial relationships and multiscale semantics, thereby achieving optimal overall performance.

In order to examine how loss weight coefficients affect the system's overall efficacy, ablation experiments on the loss function weights were performed on the Vaihingen dataset, with the results detailed in Table IV. Specifically, during the first three groups of experiments, the sum of the coefficients $\alpha_1$, $\alpha_2$, and $\alpha_3$ was fixed at 0.6. When $\alpha_1 = 0.3$, $\alpha_2 = 0.2$, and $\alpha_3 = 0.1$, the model achieved the best performance. This is because shallow features contain richer spatial structural information, which more effectively enhances the representation

TABLE IV
RESULTS OF THE ABLATION STUDY ON LOSS FUNCTION WEIGHTS

| Ablation model | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|
| Case1 | 0.1 | 0.2 | 0.3 | 91.42 | 84.53 |
| Case2 | 0.2 | 0.2 | 0.2 | 91.38 | 84.48 |
| Case3 | 0.3 | 0.2 | 0.1 | **91.57** | **84.81** |
| Case4 | 0.4 | 0.2 | 0.1 | 91.27 | 84.29 |
| Case5 | 0.5 | 0.5 | 0.5 | 91.40 | 84.51 |
| Case6 | 1 | 1 | 1 | 91.24 | 84.22 |

TABLE V
RESULTS OF THE ABLATION STUDY ON DIFFERENT DILATION RATES IN MSCA

| Dilation Rate | mF1(%) | mIoU(%) |
|---|---|---|
| (5,11,17) | 91.36 | 84.47 |
| (6,12,18) | 91.40 | 84.51 |
| (5,9,13) | **91.57** | **84.81** |

of boundary details and local structures. However, as the weight coefficients are gradually increased, the performance of the model declines, indicating that excessively large weights assigned to intermediate and shallow-layer losses may introduce additional noise and adversely affect the learning of high-level semantic features in the backbone network. Based on the ablation results, $\alpha_1 = 0.3$, $\alpha_2 = 0.2$, and $\alpha_3 = 0.1$ are selected as the optimal hyperparameter combination in this article.

To explore the influence of different dilation rates in the three intermediate branches of the MSCA module on the model's performance, ablation experiments were conducted with various dilation rate combinations. As illustrated in Table V, the model reaches its peak effectiveness—attaining 91.57% mF1 and 84.81% mIoU—under a dilation setting of (5, 9, 13). In contrast, further increasing the dilation rates does not improve semantic segmentation performance but instead leads to a slight decline. This indicates that although larger dilation rates provide a broader receptive field, overly large receptive fields may introduce noise, thus diminishing the discriminative ability of the extracted features.

To further assess the effectiveness of the proposed MSCA component in feature representation and spatial perception, a visualization analysis was conducted by comparing feature responses before and after introducing the MSCA module. The results are shown in Fig. 8. Representative vehicle and building scenes were selected to examine the impact of the MSCA module on contextual perception. As depicted in Fig. 8, feature maps without the MSCA module exhibit weak responses in target regions, and some detailed areas suffer from missed detections, indicating limited local perception capability. After incorporating the MSCA module, the network response in semantically relevant regions is significantly enhanced, and the activation range of features becomes broader. This change does not correspond to false detections; rather, it results from the multiscale convolutional structure introduced by MSCA, which enables the model to perceive broader contextual relationships and respond to neighboring regions semantically related to the target. Such enhanced feature awareness facilitates more comprehensive object segmentation in complex scenes.

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TABLE VI

RESULTS OF THE ABLATION STUDY ON DIFFERENT FEATURE MIXING STRATEGIES IN THE TRANSFORMER

| Ablation model | mF1(%) | mIoU(%) |
|---|---|---|
| w-MHA | 91.32 | 84.53 |
| w-SRA | 91.15 | 84.12 |
| w-PTM | 91.13 | 84.06 |
| SDSM | **91.57** | **84.81** |

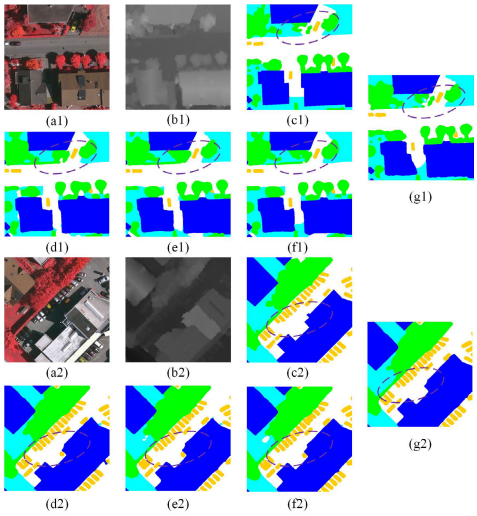



Fig. 9. Visualization results for validating the effectiveness of SDSM. (a1) and (a2) RGB images, (b1) and (b2) DSM images, (c1) and (c2) GT, (d1) and (d2) w-MHA, (e1) and (e2) w-SRA, (f1) and (f2) w-PTM, and (g1) and (g2) SAINet.

To further validate the effectiveness of the proposed SDSM module, ablation experiments employing different feature fusion strategies were conducted. The results are shown in Table VI. In these comparisons, w-MHA denotes replacing the SRE module in SDSM with the standard multihead attention in Transformer; w-SRA denotes replacing the SRE module with spatial-reduction attention (SRA) [59]; and w-PTM denotes replacing the SRE module with the pooling token mixer (PTM) [60]. SDSM refers to the proposed module. As indicated in Table VI, SDSM delivers the best performance among all schemes, with mF1 and mIoU attaining 91.57% and 84.81%, respectively, corresponding to gains of 0.25% and 0.28% compared with the standard multihead attention. This demonstrates that SDSM effectively balances local and global dependency modeling, thereby enhancing semantic segmentation performance. In contrast, w-SRA and w-PTM show slight performance degradation, indicating that simple downsampling or pooling operations are insufficient to fully capture cross-region dependencies.

To visually compare the segmentation performance of SDSM with other strategies, two samples were randomly chosen. The segmentation results of the proposed method and its variants, w-MHA, w-SRA, and w-PTM, are presented in Fig. 9. As depicted in Fig. 9, the SDSM module in SAINet more accurately distinguishes adjacent object regions while preserving boundary integrity, producing clearer object contours and segmentation results that are more consistent with the GT.

TABLE VII

RESULTS OF THE ABLATION STUDY ON THE EFFECT OF POSITIONAL ENCODING

| Type | mF1(%) | mIoU(%) |
|---|---|---|
| w/o position embedding | 90.98 | 83.80 |
| w/ position embedding | **91.57** | **84.81** |

Finally, to further investigate the dependence of SDSM on positional encoding in spatial modeling, an ablation study was conducted with and without positional encoding, while keeping the network architecture and parameter settings identical. The results are reported in Table VII, which shows that removing positional encoding leads to a decrease of 0.59% in mF1 and 1.01% in mIoU. This indicates that although SDSM enhances local contextual interaction through multi-dilation DSC, convolution operations inherently possess only relative positional sensitivity and cannot explicitly distinguish absolute spatial positions. Consequently, the absence of global positional information may result in feature misalignment. Therefore, positional encoding is retained in the SDSM architecture to provide global spatial References for the serialized features.

## V. CONCLUSION

To address the issue of intermodal inconsistency in multimodal remote sensing semantic segmentation, we present SAINet, a scale-aware and adaptive interactive semantic segmentation model tailored for RSIs. SAINet achieves an effective tradeoff between fine-grained boundary extraction and large-scale context modeling, which leads to a substantial improvement in segmentation accuracy under complex scenes. Through the AFI module, SAINet adaptively calibrates and aligns multimodal features, enabling complementary information integration while suppressing modal noise. Furthermore, the incorporation of the MSCA and SDSM modules enhances the model's capability for deep aggregation of local spatial relationships and multiscale semantics, thereby improving discrimination of small objects and large-scale structures under complex backgrounds. Although SAINet delivers competitive segmentation performance on the Vaihingen and Potsdam datasets, it still has certain limitations. First, the proposed model has a relatively large parameter count. Second, the current framework only uses two modalities for semantic segmentation. In future work, we aim to develop a more efficient and lightweight backbone network to significantly decrease model complexity without sacrificing segmentation performance, and to further expand the framework for supporting more modalities.

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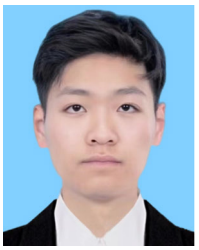
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